

FIT5202 S1 2026 Assignment 2A: Building ML Models for Road Safety Prediction (Weight: 15%)

Feedback Sheet

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			N(0%)	P(50%)	F(100%)	Comments	Marks Awarded	Total Sub Section
Part 1	Data Loading, Transformation and Exploration (45%)							
	Section Breakdown							
1. Data Loading	1.1.1	Creating a SparkContext object using SparkSession following the description in the assignment specification.	☐	☐	☒		2	37
	1.1.2	Define the schemas from metadata	☐	☐	☒		2	
	1.1.3	Using predefined schema and Load CSV	☐	☐	☒		1	
2. Data Transformation to Create Features	1.2.1	Aggregation: The dataset contains multiple rows per accident across different CSV files; group the data by collision_index to create a data frame with the necessary attributes aggregated, based on your understanding of the dataset and subsequent tasks. You will use the "severity_rating" column from the corresponding CSV (on a scale from 1-10) as your label.	☐	☐	☒		5	
	1.1.2(1)	Data imputation for missing values (1)	☐	☐	☒		5	
	1.1.2(2)	Data imputation for missing values (2)	☐	☐	☒		5	
	1.2.3(1)	Traffic density varies significantly by time of day. Extract the "Hour" from the accident timestamp and create a new categorical column, Peak_Traffic (e.g., categorising 07:00-09:00 and 16:00-18:00 as "Peak" and the others as "off-peak").	☐	☐	☒		5	
	1.2.3(2)	Please explore, identify and create another feature. This can be based on your understanding of historical data exploration from A1 or domain knowledge from the use case.	☐	☐	☒		5	
3. Data Explorartion	1.3.1	Show the statistics of the data frame.	☐	☐	☒		5	
	1.3.2(1)	Explore the dataset, plot multivariate analysis and d	☐	☒	☐	Discussion and visualization can improve te	1	
	1.3.2(2)	Explore the dataset, plot multivariate analysis and d	☐	☒	☐	Discussion and visualization can improve te	1	
Part 2	Feature Extraction and ML Training (35%)							
1. Discussion of feature selection and prepare the feature columns	2.1.1	Discuss the importance of the features and their impact to the label.	☐	☒	☐	Need more depth on label relevance.	1	28.5
	2.1.2	Write the code to create two columns based on your discussion above.	☐	☐	☒		5	
2. Preparing Spark ML Transformers/Estimators	2.2	Create two models and justify the choice.	☐	☐	☒		5	
3. Training the model	2.3.1	Write code to split the data, create pipeline, use seed=2026.	☐	☒	☐	Imputer not part of pipeline.	2.5	
	2.3.2	Discuss the metrics for model evaluation.	☐	☐	☒		5	
	2.3.3	Train the model for both pipelines, evaluate the performance and save the better model.	☐	☐	☒		10	

Part 3		Performance Tuning (20%)						
		Section Breakdown:						
Hyper-parameter Tuning and Model Optimisation	3.1	Write code for performance tuning.	☐	☐	☒		5	20
	3.2	Documenting and discussion of the preformance tuning work.	☐	☐	☒		10	
	3.3	Save the better model, full mark is awarded for any improvement (even if it's 1%).	☐		☒		5	
Additional: Quality Aspect								
Competency in memory management and data handling	Any modern laptop should be able to handle this 1.5G dataset. Mark cap applies if you choose to use less data or can't fix memory usage issues (see Ed forum).		Full Dataset is used			Mark Cap		
						100		
Code Quality	Follows programming standards, readability of the code, organization of code. Please find the PEP 8 -- Style Guide for Python Code for your reference. Here is the link: https://peps.python.org/pep-0008/		☒	☐	☐	0		0
	The output of the code should be displayed along with the code when submitting the jupyter notebook file		☒	☐	☐	0		
						Subtotal:		85.5
						Late Submission Day(s)		0.00
Final Grade						HD		85.50
Contribution to Final Marks								12.825